

Adaptive Traffic Control Using Las Vegas Algorithm

Group Number: S1_G7_ITS

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Scribe Question 1: Project System and Objective

What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Answer: Our project is addressing the issue of signal synchronisation in an intersection of the two or more roads in order to minimize the average waiting time as well as the travelling time of any vehicle. The system objective in this project is to adjust dynamically the signal Cycle Time (C_{opt}) and green phase splits depending on the real time variation in traffic.

Here the major uncertainty is the random nature of the arrival rates of vehicles. Unlike the situation in a static system where there is an assumed fixed constant arrival rate, our system deals with the randomness in the speed of vehicles such that it leads to a randomness in the vehicle arrival rate. We use Las Vegas Randomized Algorithm to select phases to manage this. The algorithm explores the uncertainty of the solution space by randomly drawing phase candidates and testing them against safety constraints in order to produce a feasible signal plan, although modified Webster and Fuzzy logic are still used to provide an approximation of the underlying traffic parameters.

Scribe Question 2: Key Random Variables and Uncertainty Modeling

Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

Answer: Our system consists of the following key random variables:

- **Vehicle Arrivals (N_t):** This is the number of vehicles arriving at the intersection in a time interval t .
- **Saturation Flow (S):** This is the maximum number of vehicles that would pass when the signal is green. Most models take this as constant, but it actually is a random variable. Every driver is different and may drive at different speeds, so the rate of flow varies.

- **Vehicle Speeds (V):** The speeds of individual vehicles before the intersection is random and variable. This variation affects directly the phase in coming on the stop line and gap acceptance logic in the simulation.
- **Queue Length (Q):** This variable refers to the number of vehicles waiting in the queue at the signal. It is a function of the vehicles arriving and the rate of flow, both of which tend to be random hence, the queue length also results to be a random variable in this system.
- **Waiting Time per Lane (W):** The time lag by vehicles in each particular lane. This is the single most important metric that we would like to minimize and varies randomly depending on signal phase allocation and queue buildup.

Uncertainty Modeling & Assumptions:

We use Webster Formula (when there are Poisson arrivals) to provide base estimation in our system. But when dealing with real time uncertainty we use Las Vegas Randomized Algorithm.

- **Algorithmic Uncertainty:** In our system we use the Las Vegas algorithm to introduce internal randomness in the runtime of the solution. It provides that the time to get a solution is random but the solution is deterministic (Error Probability = 0).
- **Traffic Uncertainty:** Fuzzy logic is applied to the continuous random variables of the traffic flow (ex: High, Low) and transforms them into discrete values that are efficiently processed by the algorithm.

Scribe Question 3: Probabilistic Reasoning and Dependencies

Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

Answer: This system is based on conditional dependence in making decisions related to control. The Las Vegas algorithm has a Generate and Verify cycle that is based on the following probabilistic relationships:

- **Conditional Feasibility (The Verification Layer):** In The Verification Layer, a phase configuration (Phase Selection) is chosen at random. This is then conditionally validated with Lane Safety Constraints, and Min/Max Timings. In case, the condition is not being satisfied (Infeasible), the system restarts to make a new random selection.
- **Inference:** It calculates difference between actual arrivals and the capacity that the actual Webster model predicts to get the Degree of Fluctuation (a derived variable).
- **Reasoning:** The reasoning engine makes use of Las Vegas property that in case a solution is given, it is known that it is correct (feasible). This system takes more time for calculation that is the calculation time will vary but we are guaranteed that the result will be safe.

Scribe Question 4: Model–Implementation Alignment

Describe how the current probabilistic model aligns with the project’s implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.

Answer:

Scribe Question 5: Cross-Milestone Consistency and Change

Describe the current state of your project’s probabilistic model. Identify the assumptions, components, or relationships that are presently well-defined, and indicate which aspects are expected to evolve or require refinement in subsequent milestones.

Answer:

Scribe Question 6: Open Issues and Responsibility Attribution

Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

Answer:

End of Submission